OOI KAI HONG

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PROFESSIONAL PROFILE

Interdisciplinary biomedical researcher with experience in cell biology, molecular and biochemical assays, advanced fluorescence microscopy, and quantitative bioimage analysis. Previously conducted research at HKUST(GZ) investigating kinase-associated regulation of microtubule nucleation and developed microscopy quantification workflows. Additional experience spans clinical-trial documentation quality control, health research ethics, systematic evidence synthesis, and peer-reviewed scientific writing. Currently pursuing a Master of Artificial Intelligence at Monash University, with a focus on strengthening computational skills and developing practical scientific software for biomedical applications.

CORE COMPETENCIES

Biomedical Research	Cell Biology | Molecular Biology | Cell Culture | Biochemistry | Immunostaining | Western Blot | Co-IP | Molecular Cloning | PCR
Microscopy & Bioimage	Confocal | Airyscan | STED | SIM | STORM | MINFLUX | Expansion Microscopy | Fiji/ImageJ | ZEN Blue | Quantitative Imaging
Clinical & Research	TMF/eTMF Quality Control | Veeva Vault | Clinical Research Documentation | Research Ethics | Systematic / Scoping Reviews | Qualitative Research | Scientific Writing
AI & Computing	Scientific Software Prototyping | Fiji/ImageJ Quantification | Python Coursework | SQL Fundamentals | Dataset Annotation | GitHub

PROFESSIONAL EXPERIENCE

Research Admin | Faculty of Medicine, University of Malaya

02/2026 – 04/2026 | Kuala Lumpur, Malaysia

- Synthesized existing literature to identify and validate research instruments for a qualitative study focused on telenursing implementation.
- Co-authored a published scoping review on tele-nursing implementation, benefits, and barriers in postnatal and maternal-child healthcare.

Research Assistant | Hong Kong University of Science & Technology (Guangzhou Campus)

09/2023 – 11/2025 | Guangzhou, China

- Investigated PI3K/PIKK-related regulation of centrosomal and acentrosomal microtubule nucleation in cultured cell models through kinase perturbation and microtubule-regrowth experiments.
- Applied confocal and super-resolution fluorescence microscopy to examine protein localization, microtubule asters and Golgi organization, including RICTOR/mTORC2-related perturbation studies.
- Performed immunostaining, cell culture, Western blot, co-immunoprecipitation, cloning and PCR to support experimental comparisons and biological interpretation.
- Developed quantitative Fiji/ImageJ workflows for ROI-based microtubule measurements and explored object-detection methods for cytosolic microtubule identification.
- Analyzed experimental imaging datasets and contributed to research documentation, figure preparation and manuscript development; microscopy research outputs remained in preparation at the end of the appointment.

Research Assistant | Faculty of Business & Economics, University of Malaya

05/2023 – 07/2023 | Kuala Lumpur, Malaysia

- Assisted with literature collection and survey-question finalization for research on financial initiatives supporting quality caregiving for older adults.
- Participated in participant recruitment and interview processes.
- Contributed to research resulting in publications in WoS-ESCI and Scopus-indexed journals.

Research Assistant | Faculty of Medicine, University of Malaya

12/2022 – 02/2023 | Kuala Lumpur, Malaysia

- Co-authored a systematic review on knee osteoarthritis interventions.

Trial Master File (TMF) Document Specialist 1 | Syneos Health Inc.

04/2021 – 10/2021 | Kuala Lumpur, Malaysia

- Conducted secondary quality control of Trial Master File documents using the Veeva Vault eTMF system.
- Met daily KPIs for document inspection and supported process efficiency.
- Collaborated with project teams to resolve document-quality issues.
- Maintained low/zero weekly error rates during document inspection.
- Received an Accountability Organization Badge.

Genetic Product Specialist Intern | Absolute Genetic Technologies

12/2020 – 02/2021 | Kuala Lumpur, Malaysia

- Interpreted genetic reports in English and Chinese.
- Assisted with survey data collection, analysis, and presentation.
- Created scientific and client-facing content and marketing materials using Canva.
- Collected biological samples to inform product development.

SCIENTIFIC SOFTWARE & APPLIED AI PROJECTS

OrganelleScope — Integrated Scientific Bioimage Analysis Desktop Application

Released Independent Research Software | 2026

- Conceptualized and guided the creation of a downloadable microscopy-analysis application using AI-assisted coding tools, informed by previous experience in biomedical research and quantitative imaging.
- Defined scientific and user requirements for consolidating two previously developed applications - Probabilistic Colocalization Estimator (PCE) and Quick Immunofluorescence Image Line Profiler (QILP) - into a unified platform.
- Iteratively reviewed application functionality and requested refinements to microscopy inspection, ROI measurements, fluorescence colocalization, line-intensity profiling and analytical reporting.

GitHub: github.com/mansonmyr/OrganelleScope

Won(da) Won — MYR/KRW Travel Decision-Support Application

Devpost Build with AI Hackathon 2026 Project | Completed and Deployed

- Conceived a web application to help Malaysian travelers monitor MYR–KRW exchange rates and coordinate currency-exchange decisions for trips to South Korea.
- Used AI-assisted development tools to translate product ideas into a functional web application with specified features including currency-rate visualizations, exchange-target simulations, regional news and collaborative Signal Rooms.
- Prepared hackathon project scope, product requirements, technical specifications and implementation checks using the Devpost Learn Skill Pack; developed for hackathon submission (award or acceptance not claimed).

GitHub: github.com/mansonmyr/Wonda_Won | Live: wondawon.netlify.app

Automated Cytosolic Microtubule Detection — YOLOv10

Biomedical Computer Vision Research Prototype | 2026

- Explored automated cytosolic microtubule detection from fluorescence microscopy images using a YOLOv10 object-detection workflow.
- Prepared microscopy imagery and LabelMe bounding-box annotations, converted training data to YOLO format and explored model training, export and Gradio-based inference.
- Reviewed detection outputs and prototype limitations as part of research-oriented computer-vision experimentation.

GitHub: github.com/mansonmyr/Automated-cytosolic-microtubule-using-object-detection-ML

Semi-Automated Microtubule Intensity Quantification — Fiji/ImageJ Macros

Scientific Image-Analysis Automation | 2026

- Created Fiji/ImageJ macro workflows for semi-automated intensity quantification from fluorescence microscopy Z-stacks.
- Used sum projection, filtering, Find Maxima, ROI generation and background correction for centrosomal and spindle-associated measurements.
- Produced structured quantitative readouts including centrosomal and spindle-pole intensity, spindle length and spindle microtubule intensity.

GitHub: github.com/mansonmyr/Semi-automated-quantification-of-in-vivo-microtubule-intensity-using-Fiji-ImageJ-macros

*Development approach: Projects combined domain expertise, scientific or product requirements, AI-assisted code generation, iterative functional review and interpretation of outputs. Software implementation was substantially AI-assisted rather than independently programmed.

RESEARCH EXPERIENCE & PROJECTS

iENGAGE Summer Student Research Programme | Wellcome-Wolfson Institute for Experimental Medicine, Queen's University Belfast

07/2023 – 08/2023

- Completed desk-based research on B-cell immune response.
- Received training in data analysis, scientific writing, project design, and experimental techniques.
- Participated in journal clubs, seminars, and a research symposium.

Health Research Ethics Practicum / Summer Research Intern | University of Malaya

07/2022 – 09/2022 | Kuala Lumpur, Malaysia

- Contributed to mHealth research for HIV prevention in partnership with Yale University and the University of Connecticut.
- Conducted transcription and data analysis for focus groups and participated in community advisory board meetings.
- Contributed to a publication in JMIR Formative Research.

Master's Thesis | University of Malaya

2022 | Kuala Lumpur, Malaysia

- Research topic: Ethical Consideration on Clinical Trials in Low-and-Middle Income Countries: an Overview of Systematic Review.
- Presented research at the SIDCER-FERCAP International Conference (2022), and University of Malaya Medical Centre Research Carnival (2023).
- Published in Discover Global Society.

Bachelor's Bioinformatics Project | Sunway University

2020 | Kuala Lumpur, Malaysia

- Investigated G-quadruplex structures in HIV-1 CRF01_AE for potential antiviral development.
- Used QGRS Mapper and NCBI BLASTn to identify potential quadruplex-forming sequences across 1,335 strains.
- Investigated potential G-quadruplex folding across multiple HIV-1 genomic regions.
- Published a narrative review based on the project.

EDUCATION

Master of Artificial Intelligence | Monash University

08/2026 – Present

- Current first-year postgraduate study in programming, computer architecture, mathematical foundations for AI/data science and IT research methods.

Master of Health Research Ethics | University of Malaya

10/2021 – 09/2022 | Kuala Lumpur, Malaysia

- CGPA: 3.61; collaborative degree programme with the Berman Institute of Bioethics, Johns Hopkins University.
- Full scholarship from the Fogarty International Centre of the United States National Institutes of Health (Grant R25TW010891).

Bachelor of Science (Hons) in Medical Biotechnology | Sunway University / Lancaster University, UK

03/2018 – 02/2021 | Malaysia

- Second Class Honours (Division One); CGPA: 3.38.

PUBLICATIONS

- Khati, A., Wickersham, J.A., Rosen, A.O., Luces, J.R.B., Copenhaver, N., Jeri-Wahrhaftig, A., Ab Halim, M.A., Azwa, I., Gautam, K., Ooi, K.H., & Shrestha, R. (2022). Ethical Issues in the Use of Smartphone Apps for HIV Prevention in Malaysia: Focus Group Study With Men Who Have Sex With Men. JMIR Formative Research, 6(12), e42939.
- Ooi, K.H., & Qi, Y. (2023). Unlocking the power of G-quadruplex for anti-HIV1 development. MEDS Clinical Medicine, 4(4), 49-52.
- Che Azmi, A., Ooi, K.H., Ab Aziz, N., Shahimi, S., & Lee, W.L. (2023). Financial Aspects of Elderly Care that Enhances their Quality of Life: A Scientometric Review. Journal of Scientometric Research, 12(3), 691-698.
- Che Azmi, A., Ooi, K.H., Shahimi, S., Ab. Aziz, N., & Silahudin, S. (2024). Financial Challenges in Music and Dance-based Interventions for Older Adults: Systematic Review. Malaysian Journal of Music, 13(1).
- Che Azmi, A., Ooi, K.H., Shahimi, S., & Ab. Aziz, N. (2025). Bibliometric analysis of the financial aspects of quality care for older people. Discover Global Society, 3(117).
- Han, L.J., Ooi, K.H., Krisna, V.B., & Phan, C.W. (2026). Ethical Challenges and Considerations in Decentralized Clinical Trial (DCTs): Insights from a Scoping Review. Therapeutic Innovation & Regulatory Science, 60, 904-924.
- Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). An umbrella review of ethical challenges in clinical trials conducted in low and middle income countries. Discover Global Society, 4(220).
- Ooi, K.H., Marina, J.R., Che, C.C., Nik Farid, N.D., & Abdullah, N.A. (2026). Implementation, Benefits, and Barriers of Tele-nursing in Postnatal and Maternal-Child Health Care: A Scoping Review. BMJ Innovations.
- Lee, J.SY., Gokilavanan, V., Ooi, K.H., & Tan, M.K.M. (2026). Beyond Lockdowns: Equity, Trust, and Social Justice in Malaysia's Pandemic Strategy. In: COVID-19 in the Pacific Rim and Beyond. Collaborative Bioethics, vol. 6. Springer, Cham.
- Lim, W.S., Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). Knowledge, Attitude, Awareness, Barriers, and Willingness to Participate in Clinical Trials: A Quantitative Survey among Undergraduate Students in Malaysia. Malaysian Journal of Medicine and Health Sciences. [Manuscript under review; not yet published]

PROFESSIONAL DEVELOPMENT & RESEARCH TRAINING

- GCI World 2026 — Matsuo-Iwasawa Laboratory, The University of Tokyo | Participant | 09/2026 – Present
- Code in Place 2026 — Stanford University | 2026
- Python Basics for Data Science — IBM | Completed
- M1 in SQL — TalentLabs | Completed
- Higher Education Course in Artificial Intelligence — ENEB Business School | Completed
- Teaching English to Foreign Learner Certificate — Foreign Talent Research Centre, MOHRSS China | Completed
- Clinical Research Associate Certificate — CCRPS | Completed
- Social and Behavioral Responsible Conduct of Research — CITI Program | Completed
- Result-based Project Management — Inter-American Development Bank | Completed

INTERNATIONAL EDUCATION & LANGUAGE DEVELOPMENT

Hanyang International Summer School | Hanyang University

06/2026 – 07/2026 | Seoul, South Korea

- Completed beginner Korean language course.

TECHNICAL & RESEARCH SKILLS

Laboratory: Cell Culture | Immunostaining | Crosslinking Co-IP | Western Blot | Sucrose Gradient Centrifugation | Molecular Cloning | PCR

Microscopy: LSM980 Confocal & Airyscan (Zeiss) | Stellaris 8 Confocal (Leica) | Axio Observer 7 | STED (Abberior STEDyCON) | SIM (Zeiss Elyra 7) | STORM (Abbelight) | MINFLUX (Abberior) | Expansion Microscopy

Flow Cytometry: BD FACSymphony A1 | FlowJo | Cell-cycle analysis

Bioimage Analysis: Fiji/ImageJ | ZEN Blue

Bioinformatics: GO & KEGG analysis | Clustal MSA | NCBI BLASTn | QGRS Mapper | TCGA

AI / Data: Python coursework and project workflows | SQL fundamentals | Dataset annotation and preparation | Object-detection experimentation | Scientific requirements and functional validation | GitHub

Research / Digital: Veeva Vault eTMF | GraphPad Prism | SPSS | Tableau | Microsoft Office | Scientific writing and evidence synthesis

LANGUAGES

- English — IELTS 7.0 (2026)
- Mandarin
- Malay
- Cantonese
- Korean — Beginner; Hanyang International Summer School (06/2026 – 07/2026)

CONFERENCE & SERVICE

Conference Support Volunteer | International Conference on Cell Organization & Genome Integrity 2024, HKUST(GZ)

12/2024 | Guangzhou, China

- Facilitated registration, session logistics, attendee support, and daily conference operations.
- Provided bilingual Chinese-English assistance to international delegates.
- Managed attendee flow to support smooth session transitions.

Oral Presenter | 22nd WHO SIDCER-FERCAP 2022, K-MEDI Hub

11/2022 | Daegu, Korea

- Presented topic on ethical consideration in clinical trial among low- and middle-income countries.